HANDIKO GESANG ANUGRAH SEJATI

IoT Firmware-Hardware & RF Engineer

Jakarta, Indonesia | handiko.gesang@gmail.com | linkedin.com/in/handiko | handiko.github.io

PROFESSIONAL SUMMARY

Innovative and detail-oriented IoT Engineer with 5+ years of professional experience specializing in firmware development, telemetry architectures, and maritime tracking solutions. Proven track record in orchestrating high-reliability hardware-software ecosystems, RF circuit design, and cost-optimized system engineering. Successfully designed and deployed national-scale Class-B AIS transmitter/receiver architectures for marine vessels and pioneered edge-computing sensor fusion algorithms for industrial agricultural automation.

PROFESSIONAL EXPERIENCE

PT. Imani Prima — (2020 – 2025)

Firmware Engineer

- Spearheaded research, firmware architecture, and hardware prototype design for Automatic Identification System (AIS) transmitters and receivers deployed by maritime and ocean-fisheries institutions.
- Architected robust embedded firmware for dual-core configurations (STM32F4 and ESP32) to process complex NMEA maritime sentences and handle real-time RF data transmission streams.
- Formulated and validated low-cost AIS Class-B transmitter electronics, solving supply-chain challenges by leveraging off-the-shelf components while ensuring compliance with stringent marine telecommunication standards.

PT. Multidaya Teknologi Nusantara (eFishery) — (2019 – 2020)

Electrical Hardware Engineer

- Engineered hardware layouts, power management configurations, and physics-based calculations for automated e-feeder units used in smart aquaculture management.
- Conceptualized and implemented a novel sensorless edge algorithm to calculate and predict fish food dispensation metrics via real-time motor electric current consumption profiling.
- Enhanced automated feeding calibration accuracy, mitigating feed waste and reducing structural hardware component costs.

PT. Integrasi Sinergi Teknologi (INSITEK) — (2018)

Engineering Intern

- Conducted analytical research on RF signal propagation profiles, determining optimal minimum structural height and multi-antenna configurations for high-altitude balloon-based Wi-Fi hotspots.

EDUCATION

Universitas Gadjah Mada (UGM) — (Graduated of 2017)

Bachelor of Engineering (B.Eng.) in Engineering Physics (GPA: 3.06)

- **Final Assignment:** Design and Development of a Multi-Mode Simultaneous Multi-Channel Modulator Based on Software Defined Radio (SDR). Applied advanced signal processing and digital modulation techniques to open-hardware RF architectures.

TECHNICAL EXPERTISE

Microcontrollers & Systems	STM32 (ARM Cortex-M), ESP32, Arduino ecosystem, Software Defined Radio (SDR)
Languages & Software	Embedded C, Embedded C++, Python, MATLAB, Altium Designer, KiCad

Hardware & RF Engineering	Circuit Design, Multi-layer PCB Layout, RF Circuits, Low Power Design (Battery Optimization), Telemetry
Protocols & Comms	AIS Marine Data, NMEA-0183, APRS (Automatic Packet Reporting System), SPI, I2C, UART, Wi-Fi, BT